

SAHIL NANDAL

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EDUCATION

Stony Brook University

Aug. 2023 – May 2027

B.S. Computer Science + Applied Mathematics and Statistics

Stony Brook, NY

- **Relevant Coursework:** Operating Systems, Database Systems, Computer Networks, Data Structures and Algorithms, Probability and Statistics

TECHNICAL SKILLS

Languages: Java, Python, C++, SQL, TypeScript, JavaScript

Frameworks and Backend: Spring Boot, Django, Flask, FastAPI, REST APIs

Databases and Infrastructure: PostgreSQL, MySQL, Redis, Kafka, Docker, GitHub Actions

Systems and Concepts: Distributed Systems, Concurrency, Fault Tolerance, System Design, Caching, Indexing

ML and Data: Vector Search (pgvector), BM25, RAG Pipelines, Model Evaluation (RAGAS), pandas, NumPy

EXPERIENCE

Software Engineering Intern – Full Stack

May 2025 – Aug. 2025

Key Events Solutions

Remote

- Optimized REST APIs and PostgreSQL schemas in Django, reducing average response latency by **40%** on data-heavy endpoints serving **10K+ concurrent users**
- Designed and enforced role-based access control (RBAC) with JWT authentication across **5+ user roles**, eliminating unauthorized access incidents in production
- Diagnosed and resolved critical production bugs via structured logging and query analysis, improving system uptime by **15%**
- Delivered **8+ features** end-to-end across Agile sprints – schema design through CI/CD deployment – on a high-volume event management platform

Software Engineering Intern

May 2024 – Aug. 2024

Cosmotech Expo India

Remote

- Architected and shipped REST APIs for ingestion, processing, and retrieval across 3 internal systems handling **500K+ daily records**
- Reduced average query latency by **30%** via targeted indexing, query rewrites, and strategic denormalization of hot-path tables
- Built a configurable data validation pipeline catching **95%+** of data integrity issues (missing fields, duplicate IDs, malformed emails, schema mismatches) before reaching downstream consumers
- Hardened system reliability through structured logging, centralized error handling, and token-based authentication across all API surfaces

PROJECTS

Lightweight SQL Query Engine | Java, Python

2026

- Built a query engine from scratch supporting SELECT, JOIN, GROUP BY, and aggregations with columnar storage and a cost-based optimizer; benchmarked **within 3x of DuckDB** on TPC-H analytical workloads
- Implemented buffer pool manager with LRU-K eviction and page-level locking; reduced average query execution time by **60%** via statistics-driven join ordering and predicate pushdown

Distributed Task Queue | Java, Redis, PostgreSQL, FastAPI

2026

- Designed a fault-tolerant job queue supporting **1M+ scheduled tasks** with priority lanes, concurrency controls, and exponential-backoff retry logic
- Achieved **5K+ tasks/sec** throughput via lock-free queue design and batched Redis pipelining; enforced exactly-once semantics via idempotency keys and transactional state machine

Distributed Key-Value Store | *Java, Spring Boot*

2025

- Built a Raft-based KV store with WAL crash recovery and log compaction; reduced recovery time by **70%** on large datasets under simulated network partitions
- Enforced strong consistency via majority-based quorum commits; maintained correctness across leader election, follower replication, and partition recovery scenarios

ThreadPool and Concurrency Library | *Java, C++*

2026

- Built a work-stealing thread pool from scratch with lock-free dequeues, achieving **2.8x speedup** over `java.util.concurrent` on CPU-bound workloads under high contention
- Implemented futures, semaphores, and barrier synchronization primitives; validated correctness via extensive stress-testing and race-condition simulation

CivicRAG: Hybrid Retrieval System | *Python, PostgreSQL, pgvector*

2024

- Combined BM25 sparse search with dense vector embeddings and cross-encoder reranking to build a hybrid document retrieval pipeline for civic records
- Reduced hallucination rate by **35%** and improved NDCG@10 by **22%** through iterative RAGAS-based evaluation and reranker tuning